# Ramanujan-Type $\pi$ Acceleration and the N-Series Geometric Constant $\pi_g$

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**Abstract**

Ramanujan's formulas for $\pi$ exhibit extraordinary acceleration through modular and hypergeometric structure. The N-Series framework approaches the same constant from a different direction: it begins with constructible polygonal approximations to the circle and accelerates them by cancelling successive even-power error terms. This note compares these two viewpoints. Ramanujan-type formulas reveal a deep analytic acceleration mechanism, while the N-Series construction defines a geometric limiting constant, denoted $\pi_g$, through chordal polygonal refinement. The comparison is complementary rather than competitive: Ramanujan's formulas give exceptional closed-form series acceleration, while the N-Series gives a constructive geometric cancellation pathway to $\pi$. Numerical order tests confirm the expected $2, 4, 6$ convergence ladder for the raw, two-scale, and three-scale polygonal approximations.

## 1 Classical Polygonal Approximation to Pi

Let

$$P_N = N \sin\Big(\frac{\pi}{N}\Big)$$

denote half the perimeter of the regular $N$-gon inscribed in the unit circle. Equivalently, $P_N$ is the polygonal chord approximation to the semicircumference of the unit circle. Since the semicircumference is $\pi$, one has

$$\lim_{N\to\infty} P_N = \pi.$$

**Geometric provenance.** Although the analytic shorthand $P_N = N\sin(\pi/N)$ contains $\pi$, the polygonal construction does not use $\pi$ as input. For the doubling scales used below, $P_N$ is constructed as one half of the perimeter of a regular inscribed $N$-gon, obtained by repeated chord bisection. The trigonometric formula is an analytic description of that constructible length.

Expanding the sine function gives

$$\sin\Big(\frac{\pi}{N}\Big) = \frac{\pi}{N} - \frac{\pi^3}{6N^3} + \frac{\pi^5}{120N^5} - \frac{\pi^7}{5040N^7} + \cdots.$$

Multiplying by $N$, we obtain

$$P_N = \pi - \frac{\pi^3}{6N^2} + \frac{\pi^5}{120N^4} - \frac{\pi^7}{5040N^6} + \cdots.$$

Thus the polygonal error has an even-power expansion:

$$P_N = \pi + \frac{C_2}{N^2} + \frac{C_4}{N^4} + \frac{C_6}{N^6} + \cdots.$$

This even-power structure is the basic target of the N-Series cancellation operators.

## 2 The Geometric Constant

The N-Series viewpoint defines $\pi_g$ as the geometric limit obtained from a scale-accelerated polygonal sequence. For a fixed integer $k > 1$, define

$$\widehat{P}_N^{(k)} = P_{kN} + \frac{P_{kN} - P_N}{k^2 - 1}.$$

Because the leading polygonal error is proportional to $N^{-2}$, this two-scale operator cancels the $N^{-2}$ term exactly.

Indeed, suppose

$$P_N = \pi + \frac{C_2}{N^2} + \frac{C_4}{N^4} + O(N^{-6}).$$

Then

$$P_{kN} = \pi + \frac{C_2}{k^2 N^2} + \frac{C_4}{k^4 N^4} + O(N^{-6}).$$

Substituting into the two-scale expression gives

$$\widehat{P}_N^{(k)} = \pi + O(N^{-4}).$$

The accelerated geometric constant is then

$$\pi_g = \lim_{N \to \infty} \widehat{P}_N^{(k)}.$$

Since $\widehat{P}_N^{(k)}$ tends to $\pi$, it follows that

$$\pi_g = \pi.$$

The significance of $\pi_g$ is not that it defines a different numerical constant. Rather, it defines $\pi$ through a purely geometric scale-cancellation process. The constant is recovered from chordal polygonal geometry, and the acceleration follows from the even-power error structure of the polygonal approximation.

## 3 Higher-Order N-Series Cancellation

The same idea extends to more scales. Using the three scales $N$, $2N$, and $4N$, define

$$\widehat{P}_N^{(3)} = \frac{1}{45} P_N - \frac{4}{9} P_{2N} + \frac{64}{45} P_{4N}.$$

The weights sum to 1:

$$\frac{1}{45} - \frac{4}{9} + \frac{64}{45} = 1.$$

They also cancel the $N^{-2}$ and $N^{-4}$ terms:

$$\frac{1}{45} - \frac{4}{9} \cdot \frac{1}{2^2} + \frac{64}{45} \cdot \frac{1}{4^2} = 0,$$

and

$$\frac{1}{45} - \frac{4}{9} \cdot \frac{1}{2^4} + \frac{64}{45} \cdot \frac{1}{4^4} = 0.$$

Consequently,

$$\widehat{P}_N^{(3)} = \pi + O(N^{-6}).$$

This is the geometric analogue of acceleration. The approximation is not improved by adding more analytic terms to a series, but by combining the same geometric construction at multiple scales so that the leading error terms cancel exactly.

## 4 Numerical Validation of the Geometric Acceleration

To verify the predicted cancellation orders, we computed the raw polygonal approximation $P_N$, the two-scale approximation $\widehat{P}_N^{(2)}$, and the three-scale approximation $\widehat{P}_N^{(3)}$ using high-precision arithmetic. For any approximation $A_N$, the observed order reported at scale $N$ is

$$p_N(A) = \log_2\left(\frac{|A_{N/2} - \pi|}{|A_N - \pi|}\right).$$

Thus each displayed row compares the errors at scales $N/2$ and $N$. The observed orders are shown in Table 1.

| $N$ | Raw order | Two-scale order | Three-scale order |
|---|---|---|---|
| 32 | 1.99791411 | 3.99875845 | 5.99923958 |
| 64 | 1.99947855 | 3.99968962 | 5.99980989 |
| 128 | 1.99986964 | 3.99992240 | 5.99995247 |
| 256 | 1.99996741 | 3.99998060 | 5.99998812 |
| 512 | 1.99999185 | 3.99999515 | 5.99999703 |
| 1024 | 1.99999796 | 3.99999879 | 5.99999926 |

Table 1: Observed convergence orders for the raw polygonal approximation, the two-scale N-Series approximation, and the three-scale N-Series approximation.

The final row gives the observed orders

$$1.99999796, \qquad 3.99999879, \qquad 5.99999926,$$

for the raw, two-scale, and three-scale approximations respectively. This confirms the predicted cancellation structure:

$$P_N - \pi = O(N^{-2}),$$
$$\widehat{P}_N^{(2)} - \pi = O(N^{-4}),$$

and

$$\widehat{P}_N^{(3)} - \pi = O(N^{-6}).$$

For completeness, the corresponding absolute errors at $N = 1024$ are

$$4.92831263353783379 \times 10^{-6},$$

$$5.79840008299119192 \times 10^{-13},$$

and

$$8.12154741379365833 \times 10^{-21}.$$

Thus the numerical data support the theoretical interpretation of $\pi_g$: the geometric constant is recovered not merely by polygonal convergence, but by a scale-cancellation ladder that removes successive even-power error terms.

## 5 Ramanujan-Type Pi Acceleration

Ramanujan's formulas for $\pi$ arise from a very different source. His series for $1/\pi$ are connected to modular equations, elliptic integrals, and hypergeometric functions. A representative Ramanujan-type structure is

$$\frac{1}{\pi} = \sum_{n=0}^{\infty} A_n,$$

where the coefficients $A_n$ contain factorial or hypergeometric factors together with rapidly decaying modular powers.

A famous example from Ramanujan's 1914 work has the form

$$\frac{1}{\pi} = \frac{2\sqrt{2}}{9801} \sum_{n=0}^{\infty} \frac{(4n)!\,(1103 + 26390n)}{(n!)^4\, 396^{4n}}.$$

This formula gains many digits per term. Its acceleration is not polygonal or chordal; it comes from the analytic and modular structure behind the series. Ramanujan acceleration is modular and analytic, whereas N-Series acceleration is geometric and scale-cancellative.

## 6 Comparison of the Two Viewpoints

The two approaches should not be treated as competing approximations in the ordinary numerical sense. Ramanujan's formula is vastly faster than raw polygonal approximation and remains one of the most remarkable analytic accelerations for $\pi$. The N-Series contribution is different: it shows that high-order acceleration can also arise from exact cancellation inside a purely geometric sequence.

The comparison is therefore structural. Ramanujan's method reveals that $\pi$ is accessible through modular and hypergeometric identities. The N-Series method reveals that $\pi$ is also accessible through constructible polygonal geometry equipped with scale-cancellation operators.

In the Ramanujan setting, the acceleration is hidden in the modular transformation. In the N-Series setting, the acceleration is visible in the moment conditions imposed on the scale weights. For the three-scale operator, these conditions are

$$\sum_{j=0}^{2} w_j = 1, \qquad \sum_{j=0}^{2} \frac{w_j}{m_j^2} = 0, \qquad \sum_{j=0}^{2} \frac{w_j}{m_j^4} = 0,$$

with scale factors

$$m_0 = 1, \qquad m_1 = 2, \qquad m_2 = 4.$$

This cancellation structure is algebraic, but it acts on a geometric sequence.

## 7 Interpretation

The N-Series does not improve Ramanujan's formula in the sense of producing a faster numerical series for $\pi$. Ramanujan's formulas are already extraordinarily efficient. Instead, the N-Series provides a complementary geometric interpretation of acceleration: the leading error terms in a constructible sequence can be removed exactly by combining multiple scales.

This distinction is important for positioning the work. The claim is not that polygonal acceleration replaces modular acceleration. The claim is that both expose hidden structure. Ramanujan

exposes hidden analytic structure in elliptic and modular functions; the N-Series exposes hidden even-power error structure in geometric approximations.

Thus the natural bridge is

$$\text{Ramanujan acceleration} \quad \longleftrightarrow \quad \text{N-Series acceleration},$$

where Ramanujan's acceleration is analytic and modular, while the N-Series acceleration is geometric and constructive.

## 8 Limitations

This note is a structural comparison, not a claim that polygonal N-Series acceleration competes numerically with Ramanujan-type series. Ramanujan's formulas are analytic, modular, and extraordinarily fast. The N-Series construction has a different purpose: it shows that exact high-order acceleration can also arise from geometric scale cancellation.

The numerical tests verify the predicted cancellation orders for the tested polygonal sequence, but they do not replace the analytic theory behind Ramanujan's formulas. The comparison is therefore complementary: one mechanism is modular and hypergeometric, while the other is geometric and scale-cancellative.

## 9 Conclusion

Ramanujan-type formulas and the N-Series construction give two distinct pathways to the same constant $\pi$. Ramanujan's pathway is analytic, modular, and hypergeometric. The N-Series pathway is geometric, chordal, and constructive. In the N-Series setting, $\pi_g$ is not a new constant but a geometric recovery of $\pi$ through scale-cancelled polygonal approximations.

The significance of this comparison is complementary. Ramanujan's formulas demonstrate the power of modular analytic acceleration, while the N-Series demonstrates that exact multi-scale cancellation can also accelerate purely geometric constructions. Together, they point to a broader principle: rapid convergence often arises when a hidden structure allows the leading error terms to be cancelled rather than merely reduced.

## References


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